

Prompt style, not translation register, drives autoformalization of Euclid into Lean 4

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Abstract

A proof in a textbook is written for people; Lean 4 is a language a computer can verify line by line. Autoformalization is the task of translating ordinary mathematical prose into Lean automatically. Using Euclid’s *Elements* Book I and GPT-4o, we ask whether the English wording of a proposition changes how well it formalizes. A counterbalanced 3×3 design (three translation registers $\times$ three prompt styles, 2,653 runs) shows that prompt style is the lever and translation register is not: prompt style spans a 24-point range in compile rate ($p < 0.001$), while translation has no significant effect ($p = 0.603$). An earlier “one translation wins” result was a prompt–translation confound. A separate probe shows a small local model had memorized the benchmark, forcing the switch to GPT-4o.

1 The question

Lean 4 is a proof language a machine can check: if a Lean proof compiles and type-checks, it is correct, with no appeal to trust. Translating informal mathematics into Lean by hand is slow expert work, so the hope is that a language model can do it—“autoformalization.” We test a narrow version of that hope on Euclid’s *Elements*, built on the **LeanEuclid** benchmark (Murphy et al., 2024).

Book I—the same 48 propositions—exists in sharply different English registers. We use three: **Heath** (1908, archaic Victorian), **Fitzpatrick** (2007, modern academic with variables named explicitly), and a plain **Modern** paraphrase. All three say the same things. Does the register change how well GPT-4o formalizes the proposition?

2 The confound, and the fix

An early run used a single prompt template whose five worked examples were all in Fitzpatrick’s style. It appeared that Fitzpatrick text formalized best, beating Heath by 8.4 percentage points ($p = 0.008$). But the Fitzpatrick translation matched the *examples in the prompt*, so register and prompt-style matching were confounded.

To separate them we crossed the two factors: three translation registers against three prompt styles (Fitzpatrick-, Heath-, and Modern-style examples), a fully crossed 3×3 design over 143 propositions (Book I plus 100 UniGeo theorems), for 2,653 total runs.

3 Result: prompt style dominates

How you prompt matters; which translation you feed in does not. Prompt style swings the compile rate across a **24-point range**—72.6% with the best style versus 48.4% with the worst ($p < 0.001$). Translation register has no significant effect once prompt style is held fixed ($p = 0.603$).

Table 1: *
Compile rate by translation $\times$ prompt style (Book I, GPT-4o)

	Fitzpatrick prompt	Heath prompt	Modern prompt
Fitzpatrick	76.3%	60.0%	45.9%
Heath	67.9%	65.5%	52.6%
Modern	73.5%	63.2%	46.8%

Read down each column (prompt fixed) and the translations barely separate; read across the columns (prompt changing) and the rates collapse. The original “Fitzpatrick wins” gap appears only in the Fitzpatrick-prompt column; under Heath-style prompts it reverses (Heath 65.5% > Fitzpatrick 60.0%), and under Modern-style prompts it reverses again (52.6% > 45.9%). On the 100 independent UniGeo theorems, Heath beat Fitzpatrick by 6.2pp—the opposite of the first claim.

4 Statistical model

A mixed-effects linear regression with proposition as a random intercept ($N = 2,653$) confirms the pattern: translation coefficients are tiny and non-significant, prompt coefficients large and significant.

	Predictor	Coefficient	SE	p
<i>Mixed-effects model coefficients</i>	Translation (Fitzpatrick vs. Heath)	+0.010	0.019	0.603
	Translation (Modern vs. Heath)	+0.009	0.021	0.665
	Prompt (Heath vs. Fitzpatrick)	−0.097	0.022	< 0.001
	Prompt (Modern vs. Fitzpatrick)	−0.242	0.024	< 0.001

Adding prompt style improves model fit by 99.5 AIC points; adding a prompt $\times$ translation interaction does not help (AIC rises by 4.3). A paired t -test on per-proposition rates ($N = 66$ items, marginalized over prompt) leaves the two translations indistinguishable: Heath 70.0% vs. Fitzpatrick 62.3%, $t(65) = -1.66$, $p = 0.102$, Cohen’s $d = -0.20$.

5 The contamination caveat

We first tried a small local model, Qwen2.5-Coder-7B (Q4_K_M), on a Jetson Orin Nano Super (8 GB, ~ 9.5 tokens/s). It had memorized the benchmark. Its *best* compile rate came from no math at all—just “Proposition N.”

	Input given	Compile rate	LeanEuclid vocab
<i>Contamination probe — Qwen2.5-Coder-7B</i>	Full Heath text	60.5%	100%
	Empty (“Proposition N”)	62.8%	100%
	Garbled vocabulary	60.5%	100%
	Shuffled clauses	60.5%	100%

The highest score from empty input means the model recites memorized structure rather than reading the math. Semantic equivalence does drop to 0% on empty input (versus 16–27% with garbled text), so it reads input to refine meaning but does not need it to produce something that compiles. GPT-4o, by contrast, is clean: 10% from empty input versus 67.9% from full text. We dropped Qwen and ran everything counterbalanced on GPT-4o.

6 Takeaway

If you want to use autoformalization on your own mathematics, the lever is not cleaning up your prose—it is styling your few-shot examples to match how you write, especially whether they introduce variables explicitly. Two methodological notes follow: a single prompt template can manufacture a spurious “wording” effect, so counterbalanced prompt designs should be standard; and the empty-input probe (does the model compile from just “Proposition N”?) is a cheap memorization check worth running before reporting autoformalization numbers.

Limits

- **One model.** The counterbalanced results are GPT-4o only; the Qwen runs are void due to contamination. Prompt sensitivity may differ elsewhere.
- **Compiling is not correctness.** Compile rates overstate real performance. On the subset semantically scored: 14% strict equivalence, 30% correct-direction. The 24-point prompt effect is measured on compilation, not correctness.
- **Uneven cells.** Fitzpatrick-prompt cells carry 430 runs each; Heath and Modern prompt cells have 133–235. The model handles the imbalance, but precision varies.
- **UniGeo, one prompt.** The 100-theorem replication used only Fitzpatrick-style prompts, so it confirms the translation non-effect but not the prompt effect.

Built on [Youklid](#) and the [LeanEuclid](#) benchmark (Murphy et al., 2024). Contamination probe ran locally on a Jetson Orin Nano Super.